

Full Length Article

Effect of oxidative atmospheres on thermochemical degradation of tobacco: Discriminating between oxidative pyrolysis and combustion

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ABSTRACT

When heated up in oxidative atmospheres, lignocellulosic materials undergo complex series/parallel reactions of pyrolysis and combustion, sometimes leading to homogeneous or heterogeneous ignition. It is therefore challenging to draw a clear border between purely degradative and oxidation reactions. This work is part of a larger investigation on the role of oxygen in the patterns, chemistry, and kinetics of tobacco's thermochemical degradation. The topic is particularly interesting for the design of innovative tobacco products that heat tobacco instead of combusting it, but it is also relevant for thermochemical conversion of all lignocellulosic biomass in general. Experiments of thermochemical degradation of tobacco have been carried out at different heating rates, under inert and oxidative atmospheres, both by Thermogravimetry (TG)/Differential Scanning Calorimetry (DSC) and in a small-scale fixed bed reactor, with contextual analysis of reaction products. In the present work, attention is devoted to the thermal effects and formation of carbon monoxide (CO) and carbon dioxide (CO₂). A methodology based on the comparison of mass loss, heat and gas release curves is applied to distinguish between reactions of oxidative pyrolysis and combustion. It is shown that below 300 °C oxidative atmospheres enhance volatiles abstraction, but heterogeneous combustion/ignition of the tobacco only occurs at temperatures above 400 °C. Molecular oxygen present in the gaseous atmosphere around the biomass particles indeed facilitates rupture of the biomass macromolecular structure during heat-up, leading to the release of some volatile components.

1. Introduction

Lignocellulosic biomass is a mixture of cellulose, hemicellulose, and lignin with relatively small portions of minerals and extractives. The patterns and kinetics of biomass pyrolysis in an inert atmosphere have been the object of extensive research and modeling efforts in the literature [1–4]. Tobacco is a particularly complex lignocellulosic biomass with relevant contents of alkaloids (especially nicotine), pectin, proteins, various sugars, organic acids, phenolic compounds, aldehydes, and aromatic hydrocarbons [5–8]; its pyrolysis has been extensively investigated in the context of the tobacco industry [9–19], as one of the reaction steps occurring during the consumption of combusted cigarettes. Due to the relevance for consumer health [20–24], attention has been devoted to the physical and reactive degradation processes [9–15], as well as detailed analysis of the resulting emissions [16–19]. Importantly, knowledge developed in the context of tobacco science can be applied to the more general context of thermochemical degradation and combustion of lignocellulosic biomass.

Cigarettes can be examined in the framework of the chemical reaction engineering. Traditional cigarettes are characterized by the formation of a hot combustion front (at the tip) with temperature levels reaching more than 900 °C [20], and a high-temperature pyrolysis and distillation zone behind the combustion front. Modern electrically heated tobacco products, which are based on controlled electric heating, the so-called Heated Tobacco Products (HTPs) [25–30], have completely changed the paradigm. Since combustion reactions are exothermic and lead to temperature increases and generation of undesired and harmful products, these devices are generally designed to heat tobacco to temperatures below around 350 °C. This operating temperature has been shown to be low enough to avoid the onset of ignition and subsequent combustion, yet sufficient for the release of low-temperature devolatilization products of tobacco. Therefore, it is of fundamental importance to understand the thermochemical degradation processes occurring at the low-temperature thermal conditions of HTPs during use and to clearly establish the thermal conditions required to avoid the combustion of tobacco for the development of novel HTPs. Discriminating

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clearly between pyrolysis and combustion upon heat up in oxidative atmospheres of a biomass, such as tobacco, is far from trivial. This especially holds true when analyzing the thermochemical degradation and influence of oxygen at the thermal conditions relevant for HTPs. As the operation of HTPs involves heating rates from around 5–10 °C/min between puffs to up to around 500 °C/min during the pre-heating phase prior to product use [29,31] different experimental techniques (e.g., Thermogravimetry (TG)/Differential Scanning Calorimetry (DSC) analysis and fixed bed reactors) under inert or oxidative atmospheres are required to cover the different heating rates and conditions involved during the thermochemical degradation of tobacco relevant to HTPs.

While it is of fundamental importance to distinguish thermochemical degradation processes from combustion in the context of HTPs, it is also very relevant in fuel science in general. Oxygen reaching the fuel surfaces during the heat-up/devolatilization stage can modify the rate and paths of thermochemical reactions, thus determining the heat release patterns and product yields, which would affect the temperature profiles in a system. These aspects are particularly critical and unexplored for oxy-combustion processes, where nitrogen is replaced by a carbon dioxide (CO₂)-rich atmosphere and oxygen concentration must be carefully tuned to guarantee the same ignition and flaming conditions as in conventional combustion. The effect of volatiles, such as CO₂, on thermochemical degradation reactions may also be relevant in the context of HTPs, in which pyrolysis of tobacco results in CO₂ emissions in an oxygen limited environment between puffs (even if the emissions are rather limited).

Previous work [18,19;32–34] emphasized that thermochemical reaction patterns and kinetics must be considered carefully if molecular oxygen from the gaseous atmosphere can reach the fuel surface during the heat-up stage. Oxygen can in some cases promote the rupture of the macromolecular structure of a solid, facilitating the abstraction of volatile matter in the early stages of heat-up, even though combustion of the solid fuel occurs in a later, separate stage. The term oxidative pyrolysis was hence used to distinguish oxygen-enhanced thermochemical degradation from combustion, where the fixed carbon in the solid fuel ignites and burns in parallel with devolatilization, leading to uncontrolled heat up and thermal degradation of the fuel. In a previous work [33], the shapes of derivative thermogravimetric (DTG) curves and carbon monoxide (CO) and CO₂ profiles from experiments in inert to oxidative atmospheres were compared to distinguish between pyrolytic, oxy-pyrolytic, and combustion processes for different solid fuels, including a bituminous coal, some plastics and two biomasses (*Robinia Pseudoacacia* and tobacco). Applying TG analysis alone to distinguish the different processes proved to be challenging for biomass because it is made up of several constituents and has a high oxygen content. CO and CO₂ are produced both from pyrolysis and combustion, but at significantly different levels, with the CO and CO₂ yields resulting from pyrolysis being only a fraction of the emissions resulting from combustion. In addition, the analysis carried out in the previous work was limited by the fact that heavy volatile species were not analyzed, and such constituents may undergo secondary reactions and even oxidation reactions in the gas phase.

In the present work, we build on previous experience and further develop the original method. A new methodology is proposed to identify regimes of pyrolysis, oxidative pyrolysis, or combustion of biomass. The methodology relies not only on experiments in TG but also on experiments in lab-scale reactors, where it is possible to sample all the products and perform a complete mass balance. The methodology will be illustrated and demonstrated for the case of tobacco.

2. Experimental

2.1. Materials

Reconstituted tobacco (RT), commonly available in tobacco shops, was used for the experiments. RT was extracted from *HEETS* tobacco

sticks (variant “Amber”) acquired from a tobacco shop in Italy. This material is made in the EU under authority of Philip Morris Products S.A. (Neuchâtel, Switzerland).

The composition of RT used in *HEETS*, known as cast-leaf tobacco, is even more complex than that of traditional tobacco, because it comprises a mixture of ground tobacco and additives [29], including approximately 15 % glycerin and minor percentages of cellulose and guar (Table S1 in Supplementary Information). Fig. 1 shows an image of the cast-leaf tobacco sample contained within a single tobacco stick.

Proximate and Ultimate analyses were performed using a LECO TG701 and a LECO CHN 628 according to the standard procedures ASTM D5142 (UNI 9903) and ASTM D 5373, respectively. The results of the analyses (average over 3–5 measures) are reported in Table 1.

Additional tests were performed on samples of guar, cellulose, and glycerin provided by Philip Morris International (PMI, Neuchâtel, Switzerland).

2.2. Preconditioning of the cast-leaf tobacco substrate

Preconditioning tests on cast-leaf tobacco samples were carried out with the scope of limiting moisture content variation. Samples were stored for at least 48 h at 22 ± 1 °C and 60 ± 2 % relative humidity (RH). Conditioning was achieved in a glass chamber with the addition of sodium nitrite salts to saturate the atmosphere. Temperature and humidity were monitored via digital sensors.

2.3. TGA tests

TGA was carried out using a Netzsch 409 TG-DSC apparatus equipped with a train of Analyzers ABB-A02020 to detect CO, CO₂, CH₄, H₂, and O₂ in the evolved gas. Samples were loaded in amounts of about 10 mg. The gas flow was controlled by Bronkhorst controllers and set at 200 mL/min. TG tests were carried out in inert conditions, using chromatographic-grade nitrogen, with air, and with O₂/N₂/CO₂ mixtures (with oxygen concentration in the range 10–30 %).

Two types of heating programs were applied in the TG:

- The sample was heated up from an ambient temperature to 110 °C with a constant heating rate (H_r , 2–40 °C/min) and held at 110 °C for 30 min to release moisture. The sample was then heated up to 750 °C with constant heating rate and held at the final temperature for 30 min.
- The sample was heated up from an ambient temperature to 750 °C with a constant H_r (2–40 °C/min) and held at this final temperature for 30 min.

2.4. Fixed bed reactor tests

Tests in either inert or oxidative atmospheres were also performed in a small quartz tubular reactor with a diameter of 2 cm and a length of 50 cm [34]. The reactor was placed in a horizontal configuration and externally heated by an electric oven, as shown in Fig. 2.



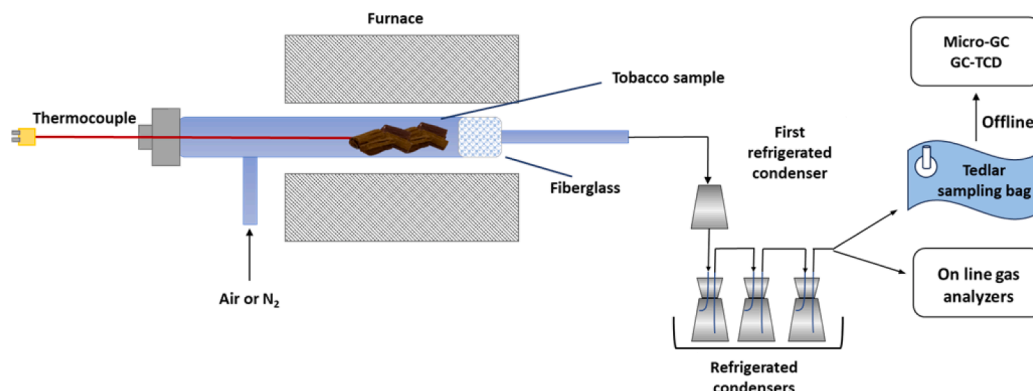
Fig. 1. Tobacco sample contained within a single tobacco stick.

Table 1

Proximate and Ultimate analysis of RT.

Moisture	Volatiles	Fixed Carbon	Ash	C*	H*	N*	O*
w%	w%	w%	w%	w% _{db}	w% _{db}	w% _{db}	(by difference) w% _{db}
13.7	67.5	10.5	8.3	42.0	6.4	1.5	41

*db = dry basis.

**Fig. 2.** Schematic representation of the experimental apparatus.

Approximately 3 g of tobacco were loaded into the reactor to carry out tests in batch-mode. Tests under inert conditions were carried out under nitrogen (N_2) flow, while tests in oxidative atmospheres were conducted either with air or with a mixture of 10 % O_2 , 40 % N_2 , and 50 % CO_2 . The gases were fed to one end of the reactor at a pre-set flow rate (20 or 70 NL/h). Thermocouples were positioned inside the sample bed to monitor temperature profiles.

Two types of experiments were carried out:

- C. The quartz reactor was positioned in the central part of a horizontal electrical oven before starting the heating program. The temperature was increased to 110 °C with a heating rate of 5 °C/min and maintained for 30 min to release moisture. The sample was then heated up to 550 °C at a constant heating rate and held there for 30 min.
- D. The quartz reactor was positioned in the electric oven after the furnace reached 550 °C, in order to obtain faster heating rates. Temperature profiles obtained within the samples were registered by thermocouples showing that the heating rates were on the order of 50 °C/min. This method is referred to as “Fast insertion” hereafter.

The experimental conditions of the different reactor tests are summarized in Table 2.

The gas flow leaving the reactor passed through a chilled impinger train made up of four refrigerated flasks at −12 °C to enhance the

Table 2

Experimental conditions of tests in the fixed bed reactor.

Test	Final temperature [°C]	Heating rate [°C/min]	Ambient reaction	Flow [NL/h]
Type C	550	5	air	20
	550	5	N_2	20
	550	5	air	70
	550	10	N_2	70
	550	10	air	70
	550	10	10 % O_2 , 40 % N_2 , 50 % CO_2	70
Type D	550	Fast insertion in the reactor at 550 °C	N_2	70
	550	Fast insertion in the reactor at 550 °C	air	70

condensation of the vapor evolved during the process.

The non-condensable gases leaving the condensation train were sent to a set of online ABB AO2020 gas analyzers to continuously monitor the concentrations of O_2 , CO , CO_2 , CH_4 , N_2O , and H_2 .

2.5. Analysis of the products

At the end of each test, the solid, liquid, and gaseous products were analyzed as follows:

- The solid substrate (char or ash) remaining after each test was weighed and analyzed with a CHN elemental analyzer for the determination of carbon (C), hydrogen (H), and nitrogen (N) content.
- The condensate recovered was weighed and characterized by Gas Chromatography-Mass Spectrometry (GC-MS) (Agilent GC 68090-MSD 5975C with DB-5 ms capillary column), Size Exclusion Chromatography (SEC) (high-performance liquid chromatography (HPLC) system HP1050 series equipped with a UV-vis diode array detector), and HPLC (Agilent 1200 Infinity series) analysis. A Karl Fischer method using a Mettler Toledo V20 Karl Fischer volumetric titrator was applied for determination of the water content. The water-free condensed product is referred to as “Liquid” hereafter.
- The gas streams were first analyzed by a system of online analyzers, with further analyses performed by collecting batches of the gas downstream of the analyzers in inert Tedlar® sampling bags, at different reaction times corresponding to different reaction temperatures. Bag contents were analyzed by means of a micro-gas chromatograph (micro-GC) Agilent 3000 and an HP5890A gas chromatograph, equipped with a 7515 Chrompack Al_2O_3/KCl capillary column and a Flame Ionization Detector (GC-FID) for the determination of light hydrocarbons (C1-C6) in addition to H_2 , CH_4 , CO , and CO_2 . An HP5700A gas chromatograph equipped with an 8700 Alltech coaxial column and a Thermal Conductivity Detector (GC-TCD) was also used for the analysis of CO_2 , CO , H_2 , and N_2 species; the gases contained in the sampling bags were analyzed by ABB NOx analyzers.

As the present paper addresses the conditions where purely pyrolytic versus oxidative regimes are established, only the evolution of main

gaseous products and the overall mass and carbon balance are disclosed. A detailed characterization of the liquid products (i.e., GC–MS, HPLC, and SEC) and minor gaseous species is planned to be reported in future papers, with a special focus on kinetics and secondary reactions.

2.6. Evolved gas analysis (EGA)

The results of the evolved gas were worked out to calculate for each species, i , in the gaseous products, its mass-based release rate, R_i , according to the equation:

$$R_i = \frac{1}{m_0} \frac{Q}{V_m} (y_i - y_{i,0}) MW_i \quad (1)$$

Where Q is the gas flow rate used in the TG/reactor experiment, m_0 the mass of the sample loaded in the experiment, y_i and $y_{i,0}$ the molar fractions of the gas products (i.e., CO, CO₂, CH₄, O₂) measured by gas analyzers at any time t and at the beginning of the test, MW_i molecular weight of the gaseous species i (g/mol), and V_m (=22.4 L/mol) is the volume of one mole of ideal gas at standard temperature and pressure.

The overall mass loss associated with the release of gaseous products was further calculated according to the equation:

$$\left(\frac{m_0 - m}{m_0} \right)_{\text{Gas}} = \frac{1}{m_0} \frac{Q}{V_m} \sum_i MW_i \int_0^{t_{\text{fin}}} (y_i - y_{i,0}) dt \quad (2)$$

and the overall rate of mass loss according to the equation:

$$\frac{d\left(\frac{m_0 - m}{m_0}\right)_{\text{Gas}}}{dt} = -\frac{1}{m_0} \frac{Q}{V_m} \sum_i MW_i (y_i - y_{i,0}) \quad (3)$$

In the TG experiments, only CO, CO₂, CH₄, and H₂ were included in Eq. (2–3). In the reactor tests, a larger array of gaseous species was monitored, including C_nH_m with 2–6 carbon atoms and molecular oxygen. It must be noted that when oxygen concentration in the gas phase decreases, its contribution to the overall integral in Eq. (3) is negative. Inclusion of oxygen in the summation of Eq. (3) guarantees that the mass balances exclude oxygen added into the system via the air (in other words, only oxygen from the fuel itself is considered in the yields).

3. Results

The following paragraph reports the results obtained from:

- 1- TG/DSC/EGA tests in inert and oxidative conditions. The mass loss, heat flow, and gas profile analyses of CO, CO₂, and CH₄ were determined to be able to discuss the differences between pyrolysis, oxidative pyrolysis, and oxidation events.

- 2- Reactor tests in inert and oxidative conditions. Temperature profiles, mass balance, and gas profile analyses of CO, CO₂, and C₂–C₆ were determined to support the TG/DSC/EGA results.
- 3- TG/DSC/EGA and reactor tests with different O₂ concentrations and CO₂-rich atmospheres. The mass loss, heat flow and evolved gas profiles (CO, CO₂, and CH₄) were compared to discuss the influence of O₂ concentration and CO₂ on the onset of pyrolysis and oxidation events.

3.1. TG measurements

3.1.1. Inert conditions

Fig. 3 depicts TG and DTG plots under inert conditions of tobacco at different heating rates, while Fig. 4 shows the corresponding evolved gas profiles (CO, CO₂, and CH₄). The results of the tests carried out in N₂ on the main additives present in RT sample, namely, guar, cellulose, and glycerin, are shown in Fig. 5.

The analysis of tests carried out under inert conditions reveals that DTG patterns are very complex due to the occurrence of multiple reactions and the presence of several constituents and additives in the cast-leaf tobacco samples.

Regarding the mass loss events, it can be observed that:

- Two mass loss events occur in the region between room temperature and around 100 °C, associated with release of moisture and bound water.
- A remarkable mass loss event is revealed by the DTG peak before 200 °C. The analysis of evolved gases does not show any peak at this temperature, suggesting that the main products at this stage are not detected by the gas analysis and are condensable at room temperature.
- A second remarkable mass loss occurs between 200 and 400 °C, with the main DTG peak at about 290 °C. The gas analysis systems detect the release of CO₂ and CO. These species cannot be regarded as combustion products because they are released in inert conditions. From Fig. 5, it can be observed that additives contribute to the mass loss in this temperature range; glycerin evaporation results in a DTG peak before 270 °C, while decomposition of guar and cellulose in DTG peaks at around 300 and 350 °C, respectively. Nicotine release also occurs in this temperature range [35].
- At even higher temperatures, three further reactive events can be observed close to 450, 500, and 550 °C. Notably, these events do not correspond to major weight losses but can be clearly appreciated by looking at the gas profiles, with CH₄ peaking at 500 °C and CO at 450 and 550 °C. Again, CO and CO₂ released in this stage result from pyrolysis reactions and cannot be regarded as combustion products because they are released in inert conditions.

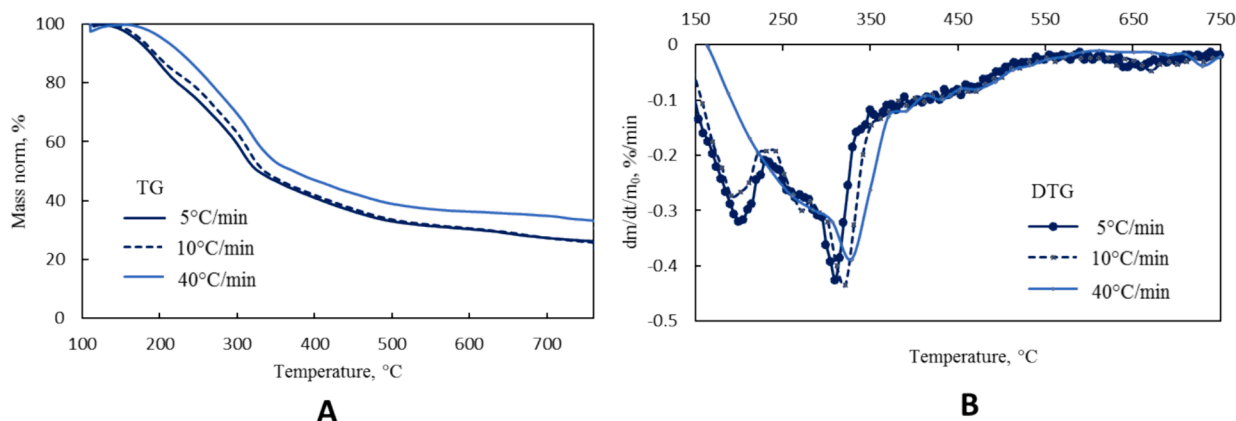


Fig. 3. Results of a thermogravimetric test on tobacco in inert conditions: A) TG, B) DTG.

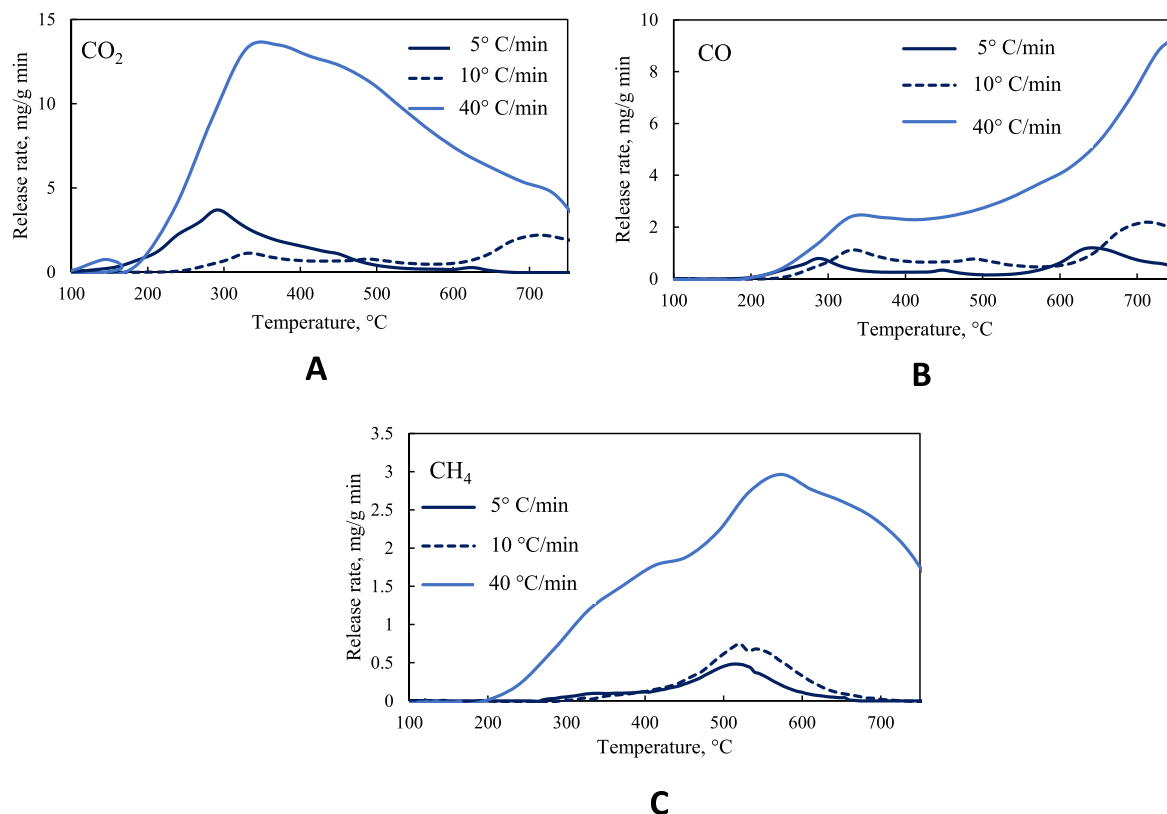


Fig. 4. Effect of heating rate on A) CO, B) CO₂, and C) CH₄ production in thermogravimetric tests of tobacco in inert conditions.

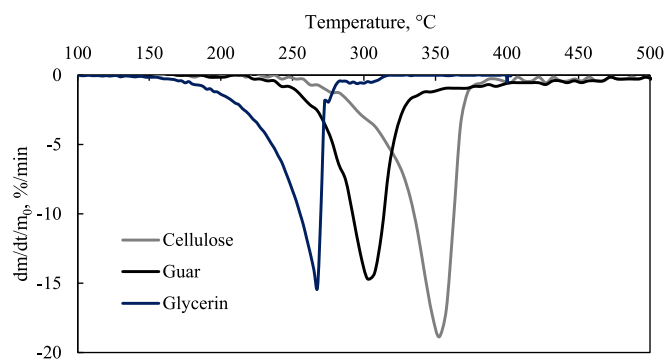


Fig. 5. DTG curves of A) guar, B) cellulose, and C) glycerin at 5 °C/min in inert conditions.

3.1.2. Oxidative conditions (air)

Figs. 6 and 7 show the results of tobacco TG tests carried out in air and compare the results with those obtained at the same heating rate under inert conditions.

It can be observed that in air:

- **TG/DTG:** Air anticipates the pyrolytic event, which under inert conditions occurs between 200 and 400 °C. Indeed, at the heating rate of 5 °C/min, the DTG peak, which occurs at 310 °C in N₂, occurs at 270 °C in air. However, the overall mass loss up to 400 °C is not increased by the oxidative conditions. At a temperature of 404–414 °C, an additional weight loss stage (~15 wt%) is observed in air for all tested heating rates. This rapid weight loss is due to a stage of fixed carbon (char) combustion.
- **DSC:** The DSC in air reveals a weak exothermic event at 290 °C and a very strong exothermic event at 410 °C. The DSC peak above 410 °C is particularly sharp and is suggestive of an ignition phenomenon.

- **Gas profiles:** The release rate of CO₂ is much higher in air than in nitrogen. Two peaks are observed in air; the second one, around 410 °C, is the most remarkable one, corresponding to combustion of the fixed carbon (char). The CO release rate is also higher in air. Interestingly, the first CO peak around 300 °C is more remarkable than the second one at 430 °C. The third CO peak, which in N₂ was observed at ~ 650 °C, is not observed in air. The main CH₄ peak, which in N₂ was observed at ~ 550 °C, in air is observed at a lower temperature of ~ 410 °C.

Regarding the effects of oxidative atmosphere on the additives, Fig. 8 shows that neither glycerin evaporation nor guar decomposition is affected, but the presence of oxygen enables a char combustion stage between 400 and 500 °C. For cellulose, decomposition appears to be anticipated (at ~ 330 °C) by the presence of oxygen and followed by a later stage of residual char combustion (at ~ 480 °C).

3.2. Reactor tests

Fig. 9A and 9B report representative results of tests carried out in the reactor with nitrogen or air at low heating rates (5 °C/min) and by fast insertion.

The temperature profiles measured by thermocouples in close contact with the sample are reported, as are the release rates of the main gaseous products measured by the online gas analyzer and micro-GC.

Analysis of the temperature profiles reveals that under inert conditions the sample temperature closely follows the furnace heating program. Conversely, the measured temperature in air exceeds the furnace temperature above 250 °C, indicating that exothermic reactions induce sample overheating.

From Fig. 9A it can be observed that at low heating rates under inert conditions, the CO and CO₂ profiles show multiple peaks between 300 and 450 °C, and CH₄ is released mainly above 500 °C. When tests are carried out in constant flow of heated air, two marked peaks of CO and

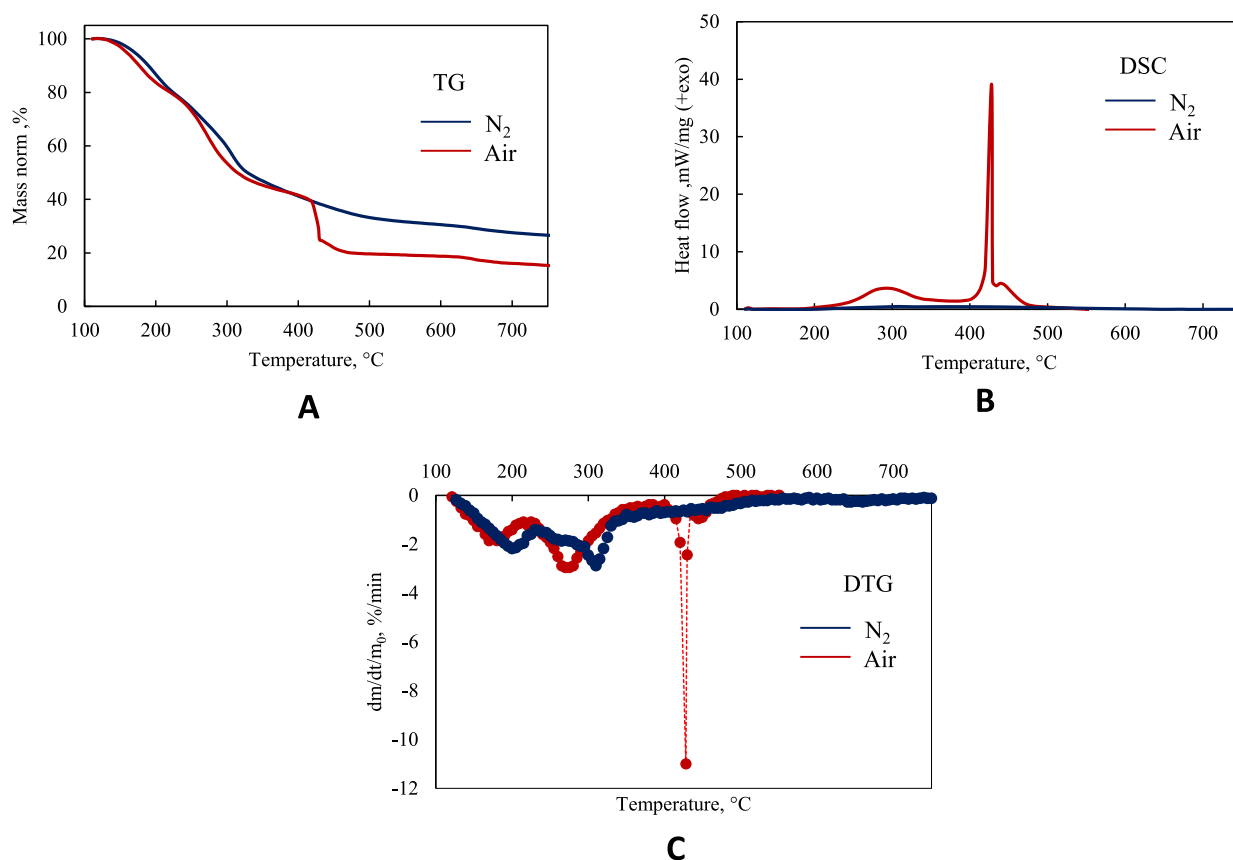


Fig. 6. A) TG, B) DTG, and C) DSC curves obtained from tobacco TG tests in air and nitrogen at 5° C/min.

CO₂ can be clearly distinguished at 270 °C and above 410 °C, when the temperature of the samples exceed the furnace temperature. The oxygen concentration decreases in parallel with the increases of CO and CO₂, suggesting the presence of gas-phase oxidation reactions. CO₂ release is distinctively higher in oxidative compared to inert conditions, but CH₄ release seems to be enhanced and anticipated in oxidative conditions. The cumulative release rates of the other C_nH_m species (with 2 to 6 carbon atoms) measured by Micro-GC in the first stages of the reaction (i.e., until $t = 30$ min, $T = 260$ °C) are comparable in oxidative and inert atmospheres; it becomes lower under oxidative conditions at temperatures above 300 °C, in correspondence with the second CO₂ peak. At higher heating rates, the temperature profiles again show sample overheating for tests carried out under a constant flow of heated air. Oxidative conditions also enhance the release rates of CO and CO₂; the peaks that were clearly separated at 5 °C/min partly overlap at the lower heating rate. Oxygen profiles confirm gaseous oxygen consumption in parallel with CO₂ formation. CH₄ release is still anticipated in oxidative atmospheres, while the release rate of gaseous hydrocarbons with 2–6 carbon atoms is distinctively lower under oxidative conditions compared to inert conditions.

Collectively, these results suggest that CO and CO₂ should be considered as pyrolysis products in tests carried out under inert conditions, while in air they also result from oxidation reactions. This hypothesis—and whether oxidation occurs at the expense of volatile products or fixed carbon—are further discussed by also examining the mass balances in the next paragraphs.

3.2.1. Mass balances for reactor tests

The mass yields in solid, liquid, and gas from the reactor tests are shown in Fig. 10. The solids yield (Y_s) includes char and ash and was obtained by recovering and weighing the solid residue in the fixed bed reactor at the end of the experiment. The gas yield (Y_g) was calculated

by integrating the results of the online gas analyzer and micro-GC. The liquid yield (Y_L) reported in the figure was obtained by calculating the difference as: $Y_L = 100 - Y_{H_2O} - Y_g - Y_s$ ($Y_{H_2O} = 13$ % from moisture is not reported in the figure). However, it must be noted that liquid yields were measured from the weight of the flasks, and the results were in reasonable agreement (± 10 %) considering the experimental weighing error.

The solid yield under inert conditions is always around 25 %. Comparison of this value with the proximate analysis of Table 1, which indicates a value of approximately 20 % for the sum of fixed carbon and ash, suggests that near-complete devolatilization of the tobacco samples in these tests. For the tests performed in air, the yield in the solid product drops to values around 10 %, close to the ash content of the tobacco (see Table 1), suggesting that combustion of fixed carbon occurred. The gas yield—and even more remarkably—the liquid yield increased moving from inert to oxidative conditions.

A mass balance on carbon was carried out considering: the yields in solid/liquid/gas reported in Fig. 10, the carbon content of the solid residue determined from proximate analysis (reported in Table S2 in the Supplementary Information), and the carbon loss associated with release of all the carbon-containing species detected by online gas analyzer and micro-GC. The distributions of carbon between solid, gas, and liquid products are shown in Fig. 11.

In the reactor tests carried out in air after heating to 550 °C, the final carbon content in the solid and liquid products decreases, while the carbon released as gaseous product increases with a faster heating rate. This suggests that oxygen in the air oxidizes the char and part of the condensable products, resulting in the formation of gaseous CO and CO₂. This will be discussed extensively in the later paragraphs.

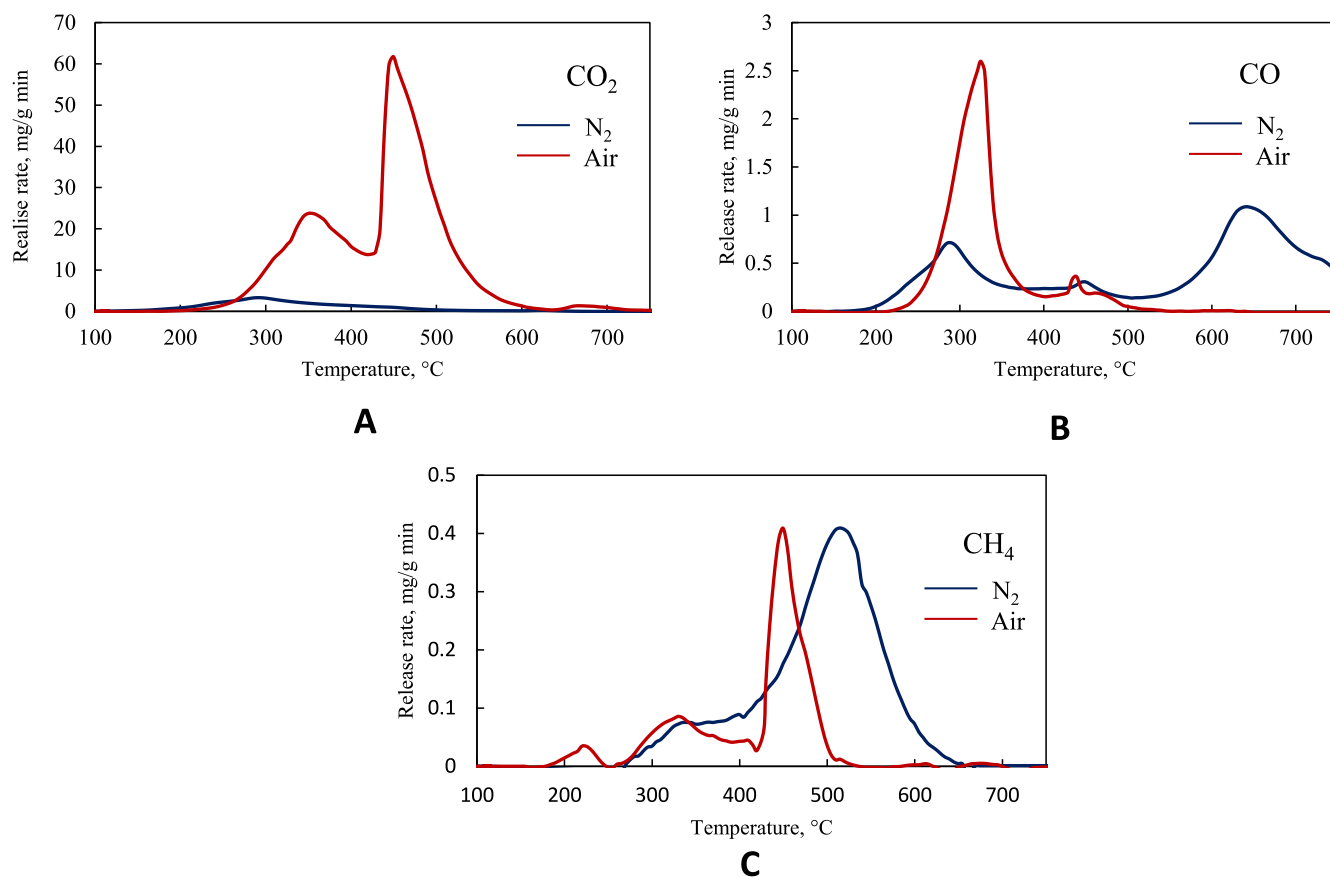


Fig. 7. A) CO_2 , B) CO , and C) CH_4 release rates of gaseous products during tobacco tests in TG at $5^\circ\text{C}/\text{min}$ in air and nitrogen.

3.3. Analysis of evolved gas

The EGA tests includes profiles of CO , CO_2 , CH_4 , and H_2 released during the TG experiments. Eq. (2–3) were applied to calculate the mass of tobacco that is transformed into CO , CO_2 , CH_4 , and H_2 throughout the TG experiments. The values of MWs to be used in Eq. (1–3) for the TG tests deserve some comments. Notably, CO and CO_2 could be either pyrolysis or oxidation products, in the case when oxidation reactions occur [33]. However, when CO and CO_2 are released as pyrolysis products, the mass loss associated to each mole of CO and CO_2 released is 28 and 44 g/mol, respectively. Therefore, the MWs of the CO and CO_2 to be used in Eq. (1–3) are 28 and 44 g/mol, respectively. The rate of mass loss curve obtained through Eq. (3) during a TG test in N_2 is reported as DTG_{gas} and are overlayed with the DTG curves obtained from the weight measurements (Fig. 12).

It can be observed that at low temperature (below 500°C) the rate of mass loss calculated from the thermobalance (DTG) is much higher than the one estimated from gas analysis (DTG_{gas}). Most of the mass loss is from volatilization of other O-HC compounds evaporated from the tobacco, including glycerin. Conversely, the DTG and DTG_{gas} profiles are in very good agreement at temperatures above 500°C .

Under oxidative conditions, different cases can be considered:

Pyrolysis hypothesis. CO and CO_2 are pyrolysis products. In this case, the mass of the tobacco sample lost with one mole of CO and CO_2 is still equal to the 28 and 44 g/mol. The MWs to be assumed in Eq. (2–3) are 28 and 44 g/mol.

Combustion hypothesis: CO and CO_2 are the result of oxidation. If oxidation reactions engage the solid carbon, only 12 g of tobacco are lost for each mole of CO and CO_2 . In this case, the MWs to be assumed in Eq. (2–3) are 12 g/mol for both CO and CO_2 .

Volatiles oxidation hypothesis: Pyrolysis generates volatiles that are then oxidized in the gas phase to produce CO and CO_2 . In this case, the

mass loss associated with each mole of CO and CO_2 is intermediate between 12 and 28/44 g/mol.

Fig. 13 compares the rate of mass loss calculated from the weight measurement (DTG) and gas analysis (DTG_{gas}) in the two extreme hypothesis of 1) pyrolysis and 2) heterogeneous combustion.

At low temperature ($T < 300^\circ\text{C}$), the DTG calculated from the weight measurement is higher than the DTG_{gas} calculated from the release of CO , CO_2 , CH_4 , and H_2 under the assumptions that CO and CO_2 are pyrolysis products and that they are oxidation products. Evaporation of glycerin and nicotine also takes place at this stage. At higher temperatures, the DTG_{gas} (grey line in Fig. 13B) calculated by working out the profiles of released CO , CO_2 , CH_4 , and H_2 (assuming that CO and CO_2 are oxidation products) shows a peak at 430°C . The same DTG peak was observed in the thermobalance (red line in Fig. 13B).

The same technique was applied to the results of reactor tests. In this case, a larger array of gaseous species was monitored, including C_nH_m with 2–6 carbon atoms and also molecular oxygen. Notably, for reactor tests, it is not necessary to formulate an a priori hypothesis of type A-B-C to assign CO and CO_2 MWs in Eq. (2–3), because the consumption of gaseous oxygen engaged in oxidation reactions is included in the overall mass balance. The onset of oxidation can, in fact, be excluded or confirmed already by looking at the oxygen concentration profiles.

Fig. 14A overlays the rate of mass loss curves estimated from the analysis of gas in the reactor tests with the DTG curves obtained in the TG experiments, carried out with similar heating rates under inert conditions. In the first stages of the reactions (temperatures below 500°C) the rate of mass loss calculated from the thermobalance (light blue) is much higher than the rate calculated from the evolved gas in the reactor tests (dark blue), suggesting that pyrolysis mainly generates hydrocarbons with $\text{C} > 2$. These are most likely condensable O-HC species, collected cumulatively as liquid [36] and include the contributions of glycerin and nicotine [35] evaporated from the tobacco.

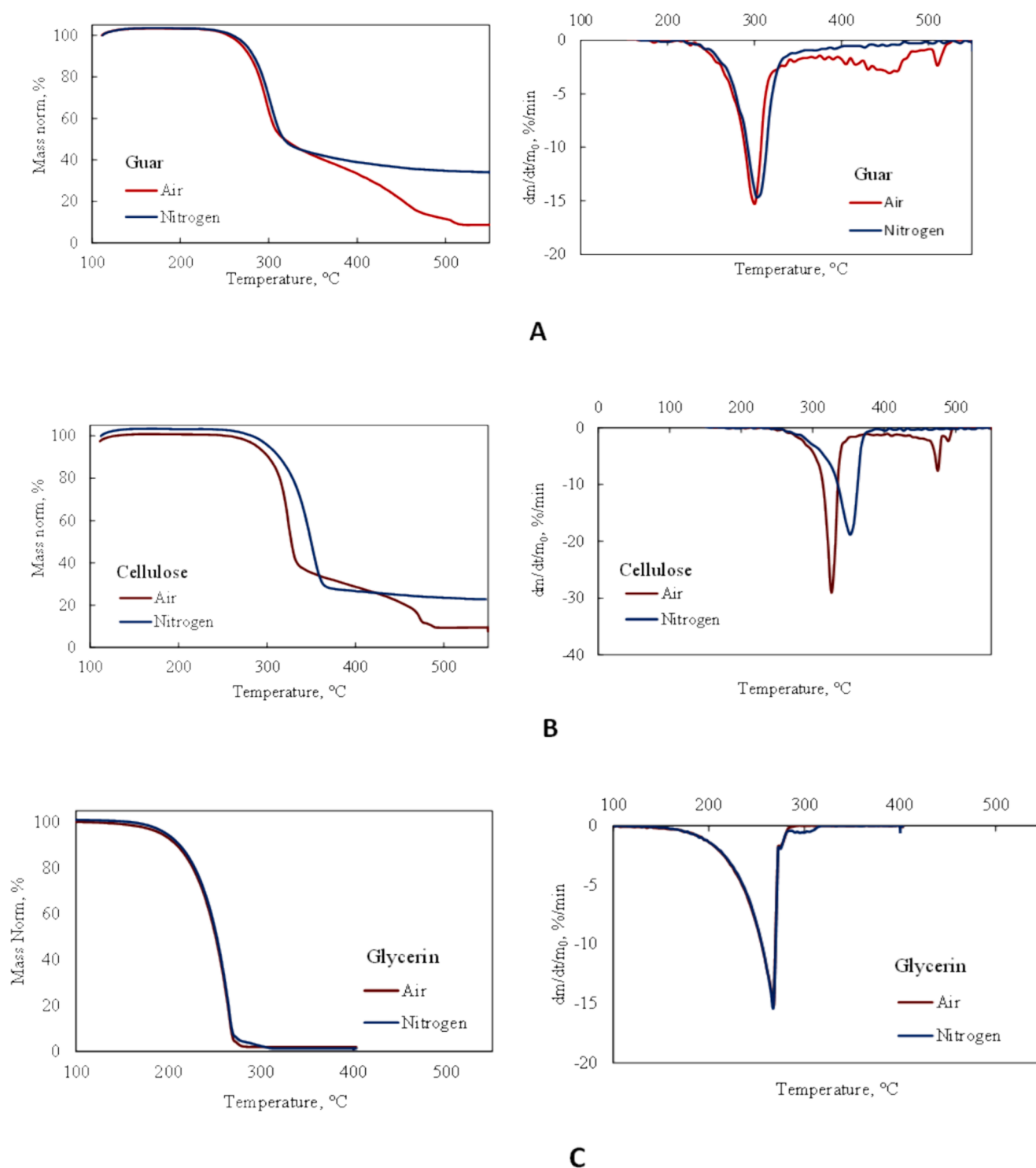


Fig. 8. TG-DTG-DSC curves of A) guar, B) cellulose, and C) glycerin at 5 °C/min in nitrogen and air.

Instead, good agreement is obtained in the later stages of the reaction, suggesting that gaseous products including CO and CO₂ prevail at higher temperatures.

Fig. 14B overlays the rate of mass loss curves estimated from the analysis of gas in the reactor tests with the DTG curves obtained in the TG experiments in air. In the first stages of the thermal degradation, when oxygen profiles are flat (Fig. 9), the rate of mass loss calculated from the thermobalance is still much higher than the rate calculated from the evolved gas in the reactor tests. This confirms that even in the presence of oxygen at low temperatures, vaporization and pyrolysis

mainly emit O-HC/HC condensable products, which are collected cumulatively as liquid products [35]. At increased temperatures, two stages where oxygen drops are observed in Fig. 9, suggesting that much of the emitted CO and CO₂ at these stages result from oxidation reactions. Comparative analysis of Fig. 14A and B suggests that in the temperature range $T = 270\text{--}400\text{ }^{\circ}\text{C}$, much of the CO and CO₂ likely result from volatile oxidation, while at high temperatures (above 400 °C), the CO and CO₂ are instead products of heterogeneous combustion of the fixed carbon (char).

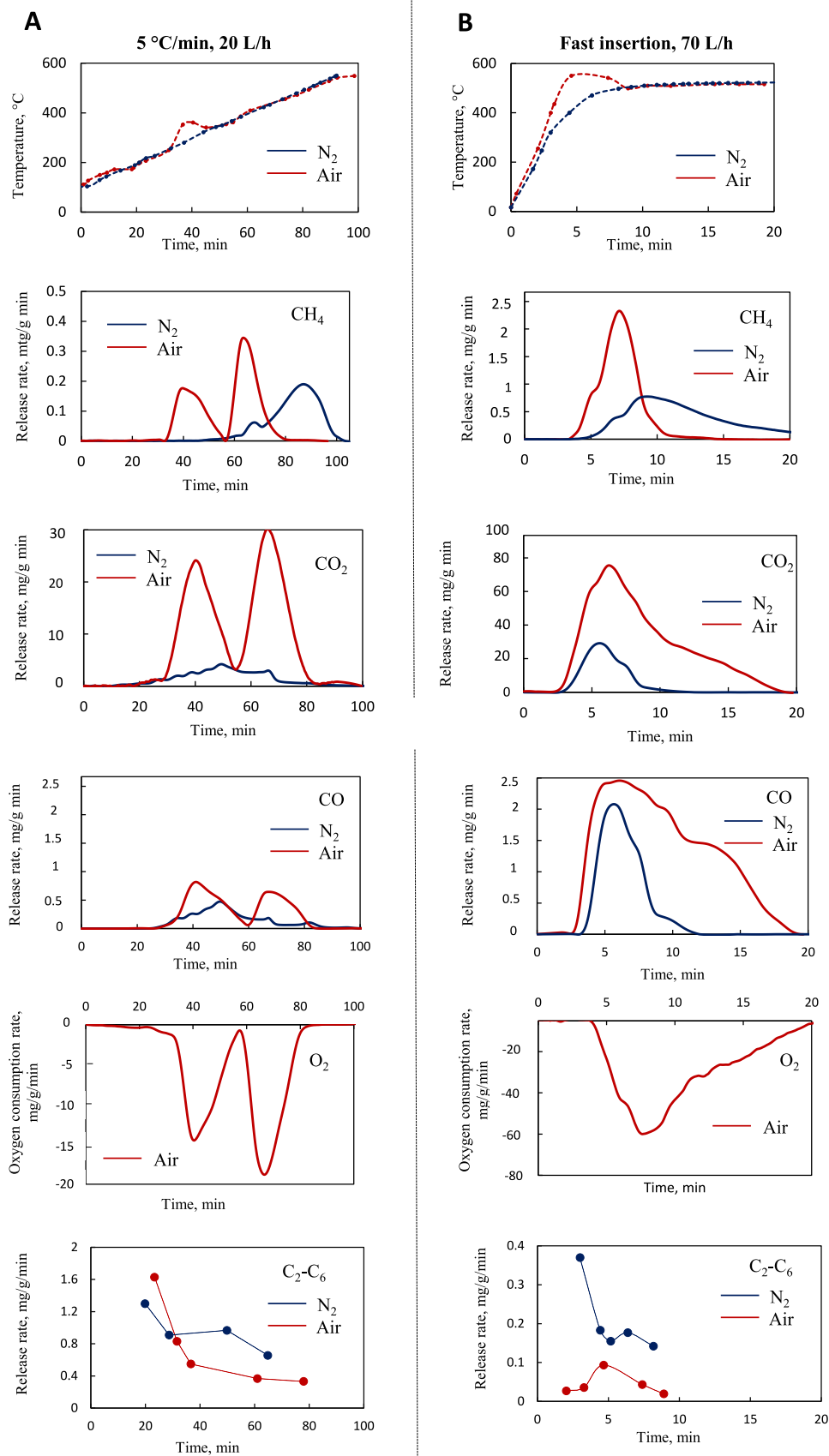


Fig. 9. A) Reactor test with 20 L/h and 5 °C/min, B) Reactor test with 70 L/h and fast sample insertion in the hot furnace.

3.4. Effect of oxygen concentration and CO₂-rich atmospheres

Fig. 15 shows the results of TG experiments with lower and higher oxygen concentration (10 % and 30 %) and with carbon oxy-combustion atmospheres (from 10–30 % oxygen in CO₂). The temperature profiles obtained in reactor tests with O₂/N₂/CO₂ atmosphere are reported in the [Supplementary Information](#).

Notably, conversion patterns do not change with oxygen concentration or upon replacement of N₂ with CO₂ in the low temperature range (up to approximately 240 °C). The second peak is sharp and is delayed when oxygen concentration is decreased and when N₂ is replaced by CO₂. The effect is even more remarkable for the last DTG peak, attributed to char combustion (above 400 °C).

The effect of gaseous atmosphere composition on the temperatures of the DTG peaks is summarized in Fig. 16. It can be concluded that with 30 % O₂ in CO₂, the effect of substitution of N₂ with CO₂ is partly—but not completely—counterbalanced by the increase of oxygen concentration, and the overall rate of conversion is lower than in air.

When N₂ is replaced by CO₂ in the reactor tests, the sample's temperature still exceeds the furnace temperature above 250 °C, as in air, but the overheating effect and temperature peaks are more modest, confirming that with 30 % O₂ in CO₂, the exothermic conversion phenomena are limited by the presence of CO₂.

4. Discriminating between thermal and oxidative processes

A logical scheme that can help discriminating between purely thermally activated processes (evaporation and pyrolysis) and oxidative phenomena is proposed in Fig. 17. It depicts the simple case of a single-component fuel, so the DTG pyrolysis pattern under inert conditions is characterized by a single peak.

In this relatively simple case, the logical path is to first compare the DTG curves obtained in inert conditions with those obtained in oxidative ones. If no differences are found, it can be concluded that oxygen in the atmosphere does not alter the course of pyrolysis.

If the DTG curve is altered in an oxidative condition, it is evident that oxygen favors the rupture of the structure through a mechanism of oxidative pyrolysis at temperatures below the ignition temperature of the material or heterogeneous combustion at temperature above the ignition temperature. To distinguish between these two possibilities, it is useful to compare the derivative mass loss obtained by the thermobalance (DTG curve) with that calculated from gas analysis (DTG_{gas}) according to Eq. (2).

Hypothesis A (CO and CO₂ are pyrolysis products) or hypothesis B (they are combustion products) was tested assuming that the MWs for CO and CO₂ of 28 g/mol and 44 g/mol, respectively, for hypothesis A, or 12 g/mol for hypothesis B as the oxygen in this case does not come from

the tobacco material.

The combination of anticipation of the DTG peaks in air, an overlap of the DTG and DTG_{gas} curves under hypothesis A, and evidence of exothermic reactions can be interpreted as a pyrolysis process where thermal feedback from volatiles oxidation determines particles overheating and accelerates the rate of the pyrolysis, without altering the nature of bond breakage and the primary pyrolysis products.

Conversely, an overlap of the DTG and DTG_{gas} curves under hypothesis B indicates the occurrence of heterogeneous reactions (above the ignition temperature). This phenomenon is also accompanied by severe exothermicity. Very sharp DSC peaks at these elevated temperatures indicate ignition and subsequent heterogeneous combustion.

Finally, if the DTG curve from the thermobalance is higher than the DTG_{gas} under hypothesis B, an oxidative pyrolysis mechanism can be assumed. This can also be accompanied by volatiles oxidation, which can be inferred by analyzing the reaction thermicity.

Application of the logical scheme proposed in Fig. 17 to the case of tobacco is not straightforward because lignocellulosic biomasses contain multiple components and the DTG curves include multiple peaks. However, the logical scheme of Fig. 17 can still be considered in discussing and interpreting the results of the present work. The main results of TG and reactor experiments are summarized in [Tables 3–4](#) for inert and oxidative conditions and discussed in subsequent sections.

4.1. Inert conditions (Table 3)

Below 400 °C, the DTG_{gas} curves are lower than the DTG calculated by TG analysis, considering the weight loss. In this temperature range, the heavy hydrocarbon species are probably collected as condensed matter and therefore not detected by the gas analyzers. Glycerin and nicotine evaporation, as well as pyrolysis of guar and cellulose, also occur at this stage. At T > 450 °C DTG_{gas} and DTG are in good agreement, suggesting that higher temperature pyrolysis generates mainly gaseous CH₄, CO, and CO₂. The absence of exothermic events both in the thermobalance (from DSC curves) and reactor (from temperature profiles) confirms that CO and CO₂ released at this stage are merely pyrolysis products and do not result from combustion.

4.2. Oxidative conditions (Table 4)

In air, at low temperatures (T < 250 °C) the DTG_{gas} curve is lower than the DTG curve calculated from the thermobalance, and the thermicity of the thermal degradation (both from the DSC and temperature profiles in the reactor) is negligible, indicating that only evaporation or pyrolysis take place in this stage, with release of O-HC/HC species not

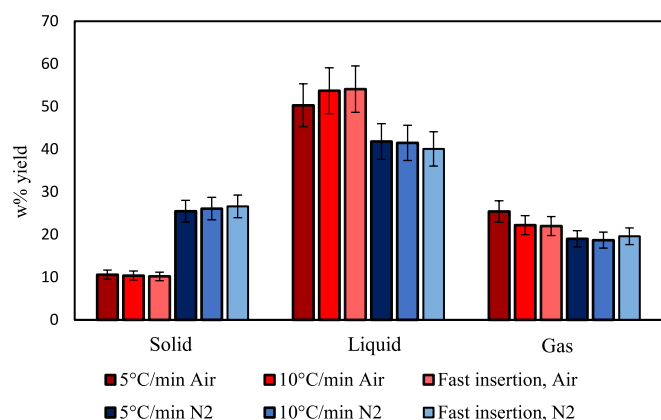


Fig. 10. Experimental solid, liquid, and gas yields from reactor tests after heating to 550 °C.

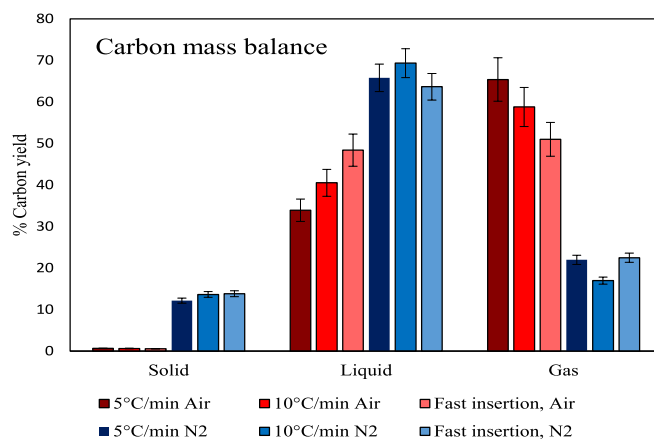


Fig. 11. Repartition of tobacco's carbon among solid, liquid, and gas products (excluding 13 % of water from tobacco's moisture) from reactor tests after heating to 550 °C.

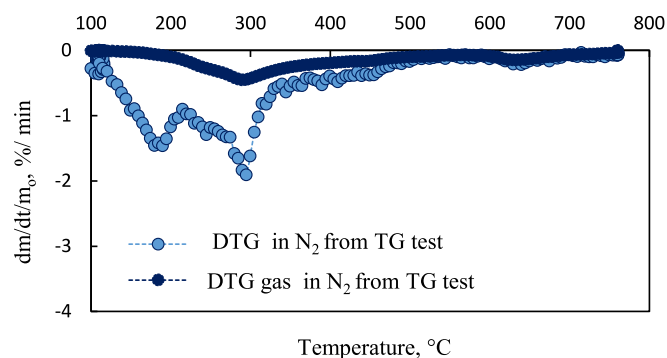


Fig. 12. Rate of mass loss calculated from the weight measurement (DTG) and gas analysis (DTG_{gas}) in a thermobalance experiment in N₂ (5 °C/min). DTG_{gas} was calculated assuming MWs of 28 and 44 g/mol for CO and CO₂, respectively.

detected by the gas analyzers (possibly condensable species cumulatively collected as liquid).

In the thermobalance, the DTG peak that occurs around 300 °C in inert conditions is anticipated in air and occurs at 270 °C. The DSC shows a modest exothermicity in this region and also in the reactor tests, with sample overheating registered above 250 °C. The question arises if at this stage, oxygen present in the atmosphere enhances bond cleavage through an oxidative pyrolysis pathway, or else enables oxidation reactions. In this region, the fact that DTG_{gas} from both TG and reactor tests are lower than the DTG from the thermobalance, demonstrates that (condensable) O-HC/HC species not detected by the gas analyzers are

also likely to be the main products in this temperature range under oxidative conditions, and only partial oxidation of volatile products occurs.

At higher temperatures, the DSC exhibits a sharp exothermic peak around 410 °C, and the reactor tests register sample overheating in parallel with increased CO and CO₂ release and higher consumption of molecular oxygen. The DTGs calculated from the thermobalance and by working out the profiles of released gases considering CO and CO₂ as combustion products are in good agreement here. All results suggest a more relevant extent of oxidation reactions in this temperature range.

The question arises here if oxidation reactions involve only the gaseous products of pyrolysis or also the remaining solid substrate and condensable products. The mass balances show that in the reactor tests under inert conditions, pyrolysis was not complete at 550 °C. Under oxidative conditions, the pyrolysis is complete and most of the fixed carbon is lost due to heterogeneous combustion. However, the carbon mass balances show that even in the oxidative tests, a large fraction of the carbon is not in the solid or gaseous products, which must be attributed to condensable O-HCs and a nonnegligible contribution of glycerin. These results suggest that under an oxidative atmosphere, pyrolysis reactions leading to heavy hydrocarbons are enhanced by molecular oxygen though oxygen-assisted bond cleavage. This conclusion is further supported by the analysis of methane profiles, which are remarkably enhanced and anticipated under oxidative conditions.

In summary, in the low temperature region where oxygen concentration profiles do not yet indicate a remarkable decay, oxidative pyrolysis anticipates the first pyrolysis stages and sample overheating is due to exothermicity of oxidative pyrolysis. At higher temperatures,

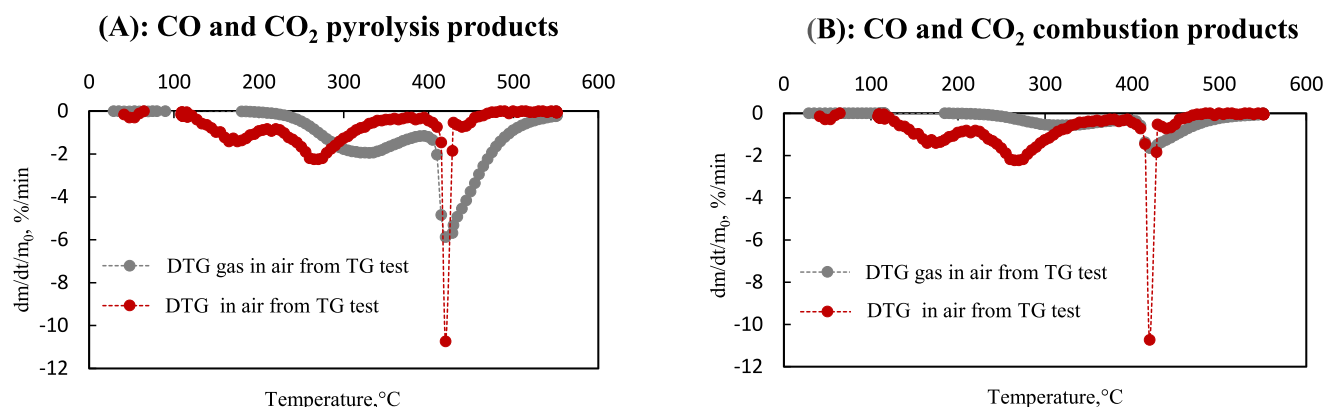


Fig. 13. Rate of mass loss mass calculated from the weight measurement (DTG) and from gas analysis (DTG_{gas}) in a thermobalance experiment in air (5 °C/min). A) DTG_{gas} calculated assuming for CO and CO₂ MWs of 28 g/mol and 44 g/mol; B) DTG_{gas} calculated assuming CO and CO₂ MW of 12 g/mol.

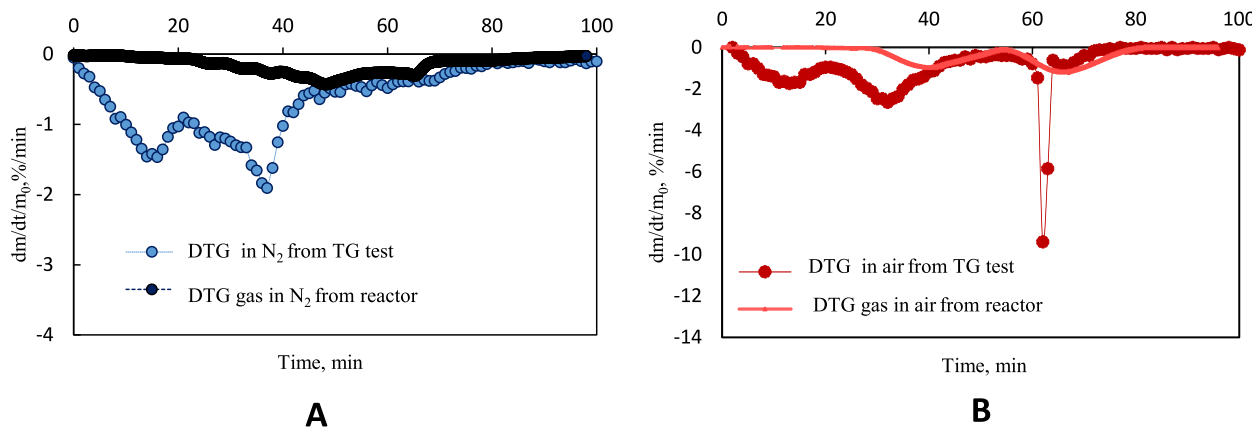


Fig. 14. Rate of mass loss mass calculated from the weight measurement (DTG) in the thermobalance and from gas analysis (Reactor) in a reactor test. A) Experiment in N₂ at 5 °C/min, (B) experiment in air at 5 °C/min.

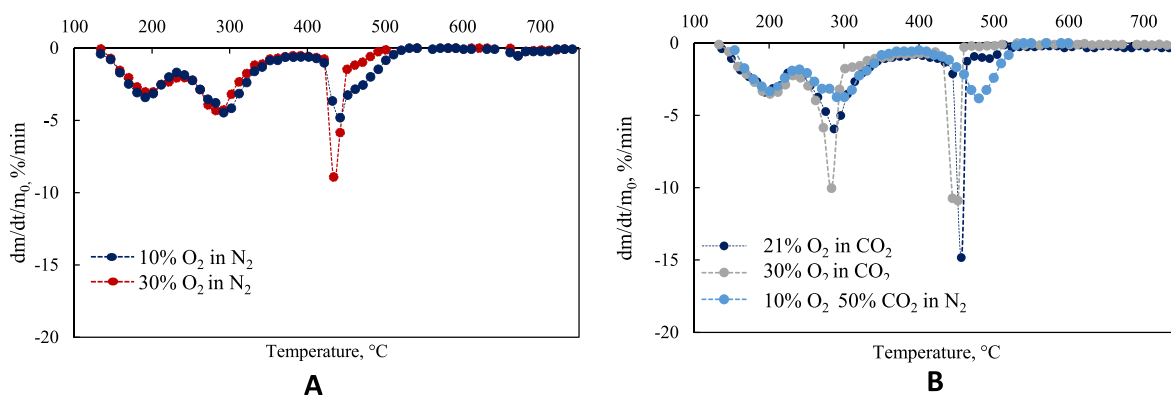


Fig. 15. DTG curves obtained from tobacco TG tests at 10° C/min with different percentages of O₂ in A) N₂ and B) CO₂.

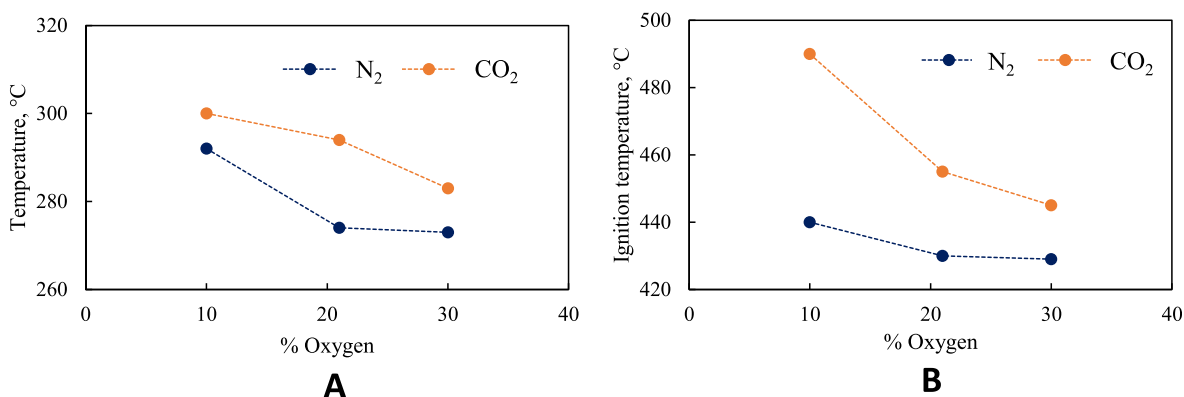


Fig. 16. DTG curves obtained from tobacco TG tests at 10 °C/min with different percentages of O₂ in N₂ and CO₂. A) 2nd DTG peak, B) Ignition temperature (3rd DTG peak).

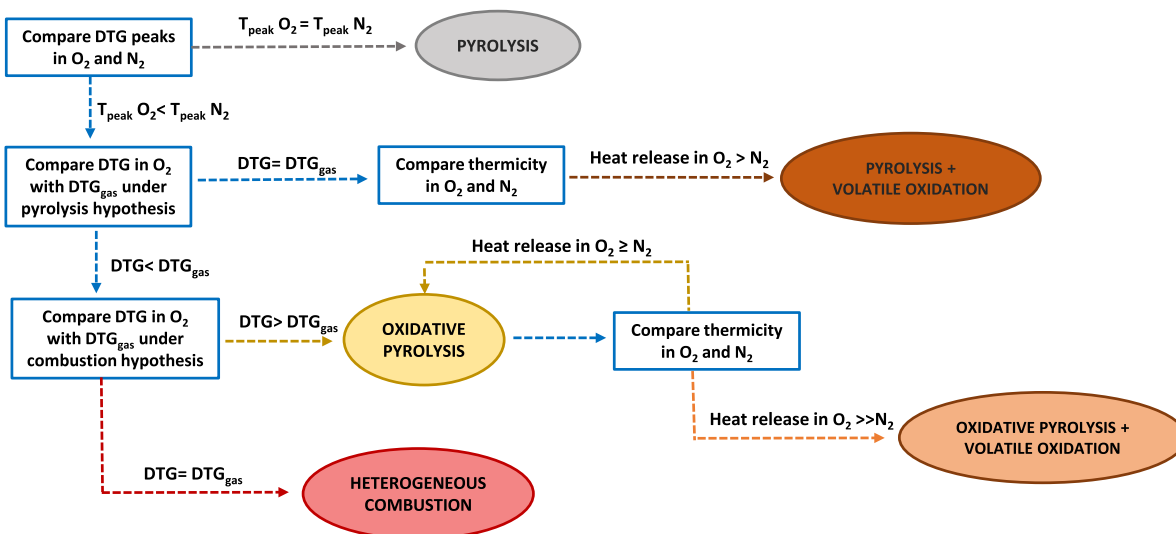


Fig. 17. Logical scheme to discuss results and discriminate between pyrolytic and oxidative phenomena.

where oxygen concentration significantly decreases and CO₂ significantly increases, oxidation of volatile pyrolysis products is more relevant, and thermal feedback from the gas phase to the samples contribute to sample overheating and further acceleration of the pyrolysis reactions.

5. Conclusions

The present work reports the results of an investigation into the role of oxygen on the patterns, chemistry, and kinetics of thermochemical degradation of reconstituted cast-leaf tobacco, typically used in heated tobacco products that are designed to heat tobacco instead of combusting it. This experimental campaign was designed to analyze the thermal degradation of tobacco under specific laboratory conditions and

Table 3

Main results in inert conditions (5–10 °C/min).

	T < 250 °C	T = 250–400 °C	T = 400–450 °C	T = 450–600 °C	T > 600 °C
Thermobalance					
DTG from EGA compared with DTG from thermobalance*	Remarkable peaks DTG gas < DTG thermobalance No peaks of CO or CO ₂	Remarkable peaks DTG gas < DTG thermobalance CO and CO ₂ remarkable peaks	Modest peaks DTG gas < DTG thermobalance CO and CO ₂ modest peaks	DTG gas = DTG thermobalance CH ₄ peak	DTG gas = DTG thermobalance CO peak
DSC	Flat				
Reactor tests					
Temperature profiles in reactor tests	Follows the heating program				
Mass loss rate from EGA in reactor tests (R _{gas}) compared with DTG from thermobalance	R _{gas} < DTG thermobalance No peaks of CO or CO ₂	R _{gas} < DTG thermobalance CO and CO ₂ remarkable peaks	R _{gas} < DTG thermobalance CO and CO ₂ modest peaks	R _{gas} = DTG thermobalance CH ₄ peak	R _{gas} = DTG thermobalance CO peak

*DTG from EGA was calculated under the hypothesis that CO and CO₂ are pyrolysis products, assuming respective MWs of 28 and 44 g/mol.**Table 4**

Main results in oxidative conditions (air 5–10 °C/min).

	T < 250 °C	T = 250–400 °C	T = 400–500 °C	T > 500 °C
Thermobalance				
DTG peaks	Remarkable peak	Remarkable peak	Remarkable peak	
EGA	No peaks of CO/CO ₂ /CH ₄	Main CO peak, Secondary CO ₂ Peak, Minor CH ₄ peak	Main CO ₂ , main CH ₄ peak, secondary CO peak,	
DSC	—	exo	exo	
DTG from EGA (pyrolysis hypothesis), compared with DTG from thermobalance	DTG _{gas} < DTG thermobalance	DTG _{gas} < DTG thermobalance	DTG _{gas} > DTG thermobalance	
DTG from EGA (combustion hypothesis), compared with DTG from thermobalance	DTG _{gas} < DTG thermobalance	DTG _{gas} < DTG thermobalance	DTG _{gas} = DTG thermobalance	
Reactor tests				
Temperature profiles	Follows heating program	Sample overheating	Sample overheating	Follows heating program
EGA	No peaks of O ₂	Negative O ₂ peaks, Main CO peak, Secondary CO ₂ peak, Secondary CH ₄ peak	Negative O ₂ peaks, Main CO ₂ peak, Main CH ₄ peak Secondary CO Peak	CH ₄ peak
Mass loss rate from EGA in reactor tests (R _{gas}) compared with DTG from thermobalance	R _{gas} < DTG thermobalance	R _{gas} < DTG thermobalance	R _{gas} = DTG thermobalance	R _{gas} = DTG thermobalance

inert and oxidative atmospheres to better understand the mechanisms behind thermochemical conversion of lignocellulosic biomass.

Experiments were carried out on tobacco samples at different heating rates up to 550 °C, in both an inert atmosphere and air, using a TG-DS and a lab-scale fixed bed quartz reactor, with contextual analysis of typical reaction products. Particular attention was devoted to thermal effects and the formation of oxygenated gaseous compounds. The results obtained with the different methods are in good agreement, and the comparison revealed three important observations:

- Multiple stages of weight loss are observed under inert conditions. Below 400 °C the main emissions are condensable O-HC/HC species, resulting from vaporization processes or pyrolysis. At higher temperatures, gaseous products including CO and CO₂ prevail.
- In oxidative conditions at low temperature, condensable O-HC/HC species resulting from vaporization and pyrolysis are still the main products emitted. The liquid yield is enhanced, and O-HC/HC species release is anticipated compared to inert conditions. Some heat release is registered between 270 and 400 °C, with contextual increase of CO and CO₂ that is attributed to partial oxidation of volatiles. Above 400 °C, a strong exothermic stage appears, which is related to heterogeneous combustion of fixed carbon (char).
- Oxidative effects above 250 °C are favored by increasing oxygen concentration in the gas mixture, while they are unfavored when N₂ is replaced by CO₂.

A logical scheme (Fig. 17) was provided for the comparative analysis of TG-DTG and EGA, with the aim of understanding the role of oxygen present in the atmosphere during thermochemical degradation. The analysis leads to the conclusion that vaporization, pyrolysis, and oxidative pyrolysis prevail at low temperatures; in other words, oxygen present in the atmosphere favors bond cleavage and anticipates the first pyrolysis stages. Above 270 °C, partial oxidation of volatiles in the constant flow of heated air contributes to sample overheating and further accelerates the pyrolysis reactions. Heterogeneous combustion occurs only above 400 °C, as consistently demonstrated by the results from the different experimental setups applied in this study.

In conclusion, the results of this work demonstrate that if reconstituted cast-leaf tobacco is heated to temperatures below 400 °C, the tobacco will not combust, and oxidation effects are restricted to the occurrence of oxidative pyrolysis and limited volatile oxidation reactions.

Future work will report a more comprehensive analysis of condensable and heavy volatile species and address the kinetics of primary and secondary reactions upon heat up of cast-leaf tobacco in inert and oxidative atmosphere.

CRedit authorship contribution statement

R. Migliaccio: Writing – original draft, Investigation. **F. Cerciello:** Writing – original draft, Investigation. **M.M. Oliano:** Writing – review & editing, Investigation, Data curation. **C. Russo:** Writing – original draft,

Investigation. **B. Apicella:** Writing – original draft, Investigation. **O. Senneca:** Writing – original draft, Supervision, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fuel.2024.132313>.

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